Academic Statement of Purpose

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I aim to support end-user programming of learners and instructors for truly personalized and self-directed learning through AI and collaborative work. To achieve my goal, I have researched human-computer interaction (HCI) and aspire to work with public schools, big EdTech platforms, and interdisciplinary researchers to extend my work to the real world at scale. UMich's versatile research infrastructure and professors can help me realize my vision. Specifically, I would like to work with **Prof. Xu Wang** on generative students and learning science research, **Prof. Anhong Guo** on accessible technologies, and **Prof. Steve Oney** on programming education tools. After graduation, I plan to stay in academia to continue my original research and foster juniors.

RESEARCH VISION

It is natural for learners and instructors to build personalized learning tools by themselves. Students create study notes to reorganize knowledge in textbooks into customized formats, and teachers prepare custom worksheets for their students. Personal learning tools integrated with digital technologies have the potential to enhance learning further. For instance, students might develop an AI agent to assist with reviewing their study notes. However, due to the high costs of creating such tools, learners and instructors are unmotivated and ill-equipped to explore their benefits.

My research vision is to realize **End-learner Programming** in which learners and instructors can actively shape their learning. I envision future learners taking the initiative in customizing existing or even creating new learning content, paths, and tools beyond given resources and classes for personalized learning. This vision aims to enable the end-user programming of diverse stakeholders in education (e.g., learners, parents, instructors) to expand learners' accessibility to and productivity in learning. For instance, I imagine a parent building a personalized textbook reader for a blind child and an adult Spanish learner making a flashcard app handy for daily commutes. My vision can personalize conventional in-person and online classrooms and ultimately grant learners the capability and autonomy to decide how and what they learn.

End-learner programming poses unique challenges and opportunities. Learners need support to reduce the **manual effort for implementation**, but learners in the same class may share or make tools together. **Learning curves for development tools** are costly, but designing a tool by oneself can be a hands-on and metacognitive learning process. Teachers are unfamiliar with **reviewing learning tools**, but they have the advantage of interacting with their target users directly and continuously. Although foundation models opened up a new paradigm for end-user programming, these challenges remain at the downstream user task levels. I will leverage the unique opportunities, AI, and HCI principles to support effective and accessible end-learner programming.

PREVIOUS AND ONGOING WORK

I have explored the challenges above to a certain extent in my research. To reduce the barrier of **high manual effort**, I explored learnersourcing, in which learners collectively generate quality data for future learners while engaging in pedagogical activities. I presented "CodeTree" [1] in CSCW'24, a programming learning platform that collects learners' self-explanations on the algorithmic steps in code examples and then aggregates them for adaptive explanations. I found that learners can collaborate to create fine-grained quality explanations at a low individual cost.

In my CHI'24 publication, I also investigated using language models as tools to **lower the learning curves** for customizing pedagogical agents. I introduced "TeachYou," [2] a system where students can learn programming by teaching a GPT4-based peer. I developed a pipeline that instructs the LLM-based peer to simulate prescribed knowledge levels. Laypeople can use natural language to configure the knowledge state of the peer, enabling them to incorporate customized teachable agents into their learning environments directly.

Recently, I explored frameworks for **reviewing pedagogical agents** during their design process. I proposed "TeachTune," [3] a tool where educators can build pedagogical agents for their students and

review the agents through automated conversations between their agents and LLM-based simulated students. This work showed instructors' need to make downstream chatbots for their students and how evaluating learning tools in breadth and depth is crucial before deploying them to actual classes.

COLLABORATION AND ENGINEERING SKILLS

The experience in my current lab at KAIST prepared me with the necessary skills to realize my vision, initiate collaborations, and expose myself to new environments. I practiced research in a big lab with more than 20 graduate students working on different research themes and methodologies. The environment encouraged me to discuss research ideas with others, find collaboration opportunities, and systematically learn to conduct research. The experience naturally prepared me to initiate collaboration outside (i.e., UMich) for my latest project [3]. To explore different research environments and expand my research skills, I also interned at UCSD. I learned qualitative research methods from my internship projects [4, 5] and applied thematic coding and formative interviews in my subsequent research [2]. I also worked at a startup company as a software engineer for two years and made applications that served more than a million users. I not only learned how to design scalable software but also observed a successful instance of end-user programming where engineers create tools to improve their developer experience. I am comfortable creating prototypes in a day and have built all frontend, backend, and data processing pipelines for my projects.

MOTIVATION AND CAREER PLAN

Through my education and volunteer work, I have engaged with Korean, Western, and African learning environments, gaining valuable insights into how student agency and resource accessibility can enhance engagement. I want to help future learners realize how much autonomy and creativity they can have in their learning and enjoy them as I did. I aim to enable this vision through system building to impact users directly. Having explored both academia and industry, I believe that UMich offers a unique platform to maximize the impact; it provides the freedom to collaborate with schools, companies, and researchers, enabling a broader impact and innovation than industry alone.

I highly value opportunities to make real-world impacts from my research. My experience preparing for an eight-million-dollar EdTech competition, which emphasized accessible learning to developing countries and practical application, encouraged me to consider how my research can deliver to and impact the real world. I learned that UMich's Human-AI lab deployed their ImageExplorer app to 300+ users with disabilities in the wild. This is the scale of impact I want to deliver throughout my Ph.D. program. UMich's community interest in human learning, the pursuit of practical values, and extensive networks with both academia and industry will make these opportunities more accessible, helping me step up to real-world problems beyond research in a lab.

After graduation, I plan to stay in academia, where I can have the merit of interacting with junior students. I enjoyed consulting 50 undergraduate students for six months as a department counseling assistant and mentoring six undergraduate interns in my research projects. I realized my experience and talent could help juniors in my community explore career paths and practice research, as my seniors helped me. As a first-generation college student, all the community support during my college life helped me find my career. I want to contribute back by fostering scholars and academic culture.

[1] **Hyoungwook Jin** and Juho Kim. 2024. CodeTree: A System for Learnersourcing Subgoal Hierarchies in Code Examples. Proc. ACM Hum.-Comput. Interact. 8, CSCW1, Article 31 (April 2024).

[2] **Hyoungwook Jin**, Seonghee Lee, Hyungyu Shin, and Juho Kim. 2024. Teach AI How to Code: Using Large Language Models as Teachable Agents for Programming Education. In Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI '24).

[3] **Hyoungwook Jin**, Minju Yoo, Jeongeon Park, Yokyung Lee, Xu Wang, and Juho Kim. TeachTune: Reviewing Pedagogical Agents Against Diverse Student Profiles with Simulated Students. arXiv preprint arXiv:2410.04078 (2024). *(Under Review)*

[4] Jane L. E, Yu-Chun Grace Yen, Isabelle Yan Pan, Grace Lin, Mingyi Li, **Hyoungwook Jin**, Mengyi Chen, Haijun Xia, and Steven P. Dow. 2024. When to Give Feedback: Exploring Tradeoffs in the Timing of Design Feedback. In Proceedings of the 16th Conference on Creativity & Cognition (C&C '24).

[5] Yu-Chun Grace Yen, Jane L. E, **Hyoungwook Jin**, Mingyi Li, Grace Lin, Isabelle Yan Pan, and Steven P. Dow. 2024. ProcessGallery: Contrasting Early and Late Iterations for Design Principle Learning. Proc. ACM Hum.-Comput. Interact. 8, CSCW1, Article 112 (April 2024).